Ye Wang

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Research Interest

My research primarily focuses on Physical AI systems and their security. Specifically, I investigate both offensive and defensive techniques for IoT, CPS, and AI-enabled systems. My research interests include:

- Trustworthy and Privacy-preserving ML/AI: Adversarial ML; Prompt Injection; AI Misinformation and Misuse.
- Physical AI: IoT/CPS; Mobile; Wireless Systems; Side and Covert Channels.
- Communication Security: NFC; Optical Fiber Communication; GPS Spoofing.

Education

- **University of Kansas** — *Lawrence, Kansas, United States*
  Ph.D. in Computer Science, Department of Electrical Engineering and Computer Science — Jan. 2020 – May. 2026
  Advisor: Prof. Fengjun Li & Prof. Bo Luo
- **Beihang University** — *Beijing, China*
  M.Eng. in Optical Engineering, School of Instrumentation Science and Optoelectronic Engineering — Sep. 2011 – Mar. 2014
  Advisor: Prof. Xiaxiao Wang
- **Beihang University** — *Beijing, China*
  B.S. in Electronic Engineering, School of Instrumentation Science and Optoelectronic Engineering — Sep. 2007 – Jul. 2011
  Advisor: Prof. Zhongyi Chu

Publications

Students mentored by me Conference Papers

[C1] Ren, Liangqin and Liu, Zeyan and **Wang, Ye** and Chen, Yuxin and Li, Fengjun and Luo, Bo. PhantomSeal: Proactive Deepfakes Defense with Identity/Context Protection and Forensic Tracing. *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (**CCS**), 2026. Accepted*

[C2] **Wang, Ye** and Li, Yuying and Luo, Bo and Li, Fengjun. When Side Channels Meet Covert Channels: A Practical Fully Sensor-Driven Attack Chain. *International Conference on Information and Communications Security (**ICICS**) 2026. Accepted*

[C3] **Wang, Ye** and Luo, Bo and Li, Fengjun. Beyond Conventional Triggers: Auto-Contextualized Covert Triggers for Android Logic Bombs. *The Network and Distributed System Security (**NDSS**) Symposium 2026.*

[C4] **Wang, Ye** and Luo, Bo and Li, Fengjun. A Novel Fully Sensor-driven Attack Chain. *Annual Computer Security Applications Conference (**ACSAC**) 2025, Poster Session*

[C5] **Wang, Ye** and Liu, Zeyan and Luo, Bo and Hui, Rongqing and Li, Fengjun. The Invisible Polyjuice Potion: an Effective Physical Adversarial Attack against Face Recognition. *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (**CCS**)*, 2024, 3346–3360.

[C6] Li, Kevin and Wang, Zhaohui and **Wang, Ye** and Luo, Bo and Li, Fengjun. Poster: ethics of computer security and privacy research-trends and standards from a data perspective. *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (**CCS**), Poster Session* 2023, 3558–3560.

[C7] Fan, Wei and Huang, Weiqing and Zhang, Zhujun and **Wang, Ye** and Sun, Degang. A Near Field Communication (NFC) security model based on the OSI reference model. *2015 IEEE Trustcom/BigDataSE/ISPA*, 2015, 1324–1328.

Journal Papers

[J1] Kong, Qingshan and Kang, Di and **Wang, Ye** and Zhang, Meng and Huang, Weiqing. Eavesdropping Attacks on Optical Fiber Communication and Countermeasure of Optical Fiber Sensing Technology. *Journal of Information Security Research*, 2.2 (2016): 123.

[J2] Wang, Xiaxiao and **Wang, Ye*** and Qin, Yi and Yu, Jia. Ratio error of all fiber optical current transformer caused by mean wavelength's fluctuation. *Infrared and Laser Engineering*, 2015, 44.1 (2015): 233-238.

[J3] Wang, Xiaxiao and Wang, Xichen and **Wang, Ye** and Feng, Xiujuan. A novel Faraday effect-based semi-physical simulation method for bandwidth of fiber-optic gyroscope. *Optik*, 2014, 1358–1360.

[J4] Wang, Xiaxiao and **Wang, Ye*** and Wang, Xichen and Wang, Aimin and Peng, Zhiqiang. Experimental research on the dynamic characteristics of fiber-optical current transformer. *Power System Protection and Control*, 42.3 (2014): 9-14.

[J5] Wang, Xiaxiao and Qin, Yi and **Wang, Ye***. Errors of fiber delay line polarization crosstalk for all fiber optical current sensors. *Optics and Precision Engineering*, 22.11 (2014): 2930-2936.

[J6] Wang, Xiaxiao and **Wang, Ye*** and Li, Chuansheng and others. Measurement method and experimental research of the temperature dependence of the phase delay of quarter-wave plates. *Chinese J Lasers*, 40.12 (2013): 1205004.

RESEARCH AND PROFESSIONAL EXPERIENCE

- **Institute for Information Sciences, University of Kansas** — *Lawrence, Kansas, United States*
  Graduate Research Assistant — Jan. 2020 – Present
  - Non-intrusive physical-layer masking for preventing side-channel leaks via accelerometers.
  - Combining motion-sensor side channels with covert vibration channels to form a practical attack chain.
  - Stealthy sensor-enabled logic bombs for Android that evade static analysis, dynamic analysis, and user awareness.
  - Proactive deepfakes face swap defense with identity/context protection and forensic tracing.
  - Develop an effective physical adversarial attack against face recognition CNN models.

- **Institute of Information Engineering, Chinese Academy of Sciences** — *Beijing, China*
  Assistant Research Fellow (full-time) — Mar. 2014 – Dec. 2019
  - Designed and implemented GPS spoofing detection techniques for UAV navigation.
  - Developed a fiber-optic communication system security monitoring framework, focusing on intrusion detection.
  - Developed an unauthorized recording device detection system based on weak magnetic signal analysis.
  - Conducted research on security modeling for Near-Field Communication (NFC) devices.

- **Institute of Optoelectronics Technology, Beihang University** — *Beijing, China*
  Graduate Research Assistant — Sep. 2011 – Mar. 2014
  - Conducted research to improve the dynamic response and measurement accuracy of fiber-optic current transformers.

HONORS AND AWARDS

- NDSS Fellowship – Internet Society — 2026
- Doctoral Student Research Fund Award – University of Kansas (KU) — 2025
- Graduate Engineering Association Award – KU — 2024, 2025
- DAVID D. and MILDRED H. ROBB AWARD – EECS, KU — 2024, 2025
- ACM CCS Travel Grant Award – ACM SIGSAC, NSF — 2024
- Graduate Student Travel Fund – KU Student Senate — 2024, 2025
- CANSec Workshop Travel Grant Award – CANSec, NSF — 2022, 2024, 2025
- The second prize of the Science and Technology Award – Ministry of Industry and Information Technology, PRC — 2019
- Annual Outstanding Researcher Award – Institute of Information Engineering, CAS — 2016, 2018
- Outstanding Master's Thesis Award – Beihang University — 2014
- Science and Technology Award – Ministry of Education, PRC — 2013
- Graduate Guanghua Scholarship – Beihang University — 2013
- Outstanding Undergraduate Thesis Award – Beihang University — 2011

TEACHING EXPERIENCE

- **Graduate Teaching Assistant**
  University of Kansas
  - EECS 569: Computer Forensics — Fall 2024
    Instructor: *Dr. Bo Luo*
  - EECS 565: Introduction to Information and Computer Security — Fall 2025, 2024, 2023
    Instructor: *Dr. Fengjun Li*
  - EECS 447: Introduction to Database Systems — Spring 2026, 2024, 2023
    Instructor: *Dr. Bo Luo*
  - EECS 268: Programming II — Spring 2025
    Instructor: *Dr. John Gibbons*
- **Teaching Assistant**
  University of Chinese Academy of Sciences
  - Physical Space Information Security — Spring, 2017
    Instructor: *Prof. Degang Sun*

MENTORING EXPERIENCE

1. Liangqin Ren, PhD candidate, The University of Kansas, 01/2025-present
   Project: Proactive Deepfakes Defense with Identity/Context Protection and Forensic Tracing, CCS 2026
2. Yuying Li, PhD student, The University of Kansas, 01/2025-present
   Project: A novel attack chain to improve the practicality of side channel attacks, ICICS 2026
3. Weihang Hu, master student, University of Chinese Academy of Sciences, 9/2018-9/2019
   Project: CNN-based electromagnetic spectrum analysis (Master's thesis)

Conference Presentation

- Beyond Conventional Triggers: Auto-Contextualized Covert Triggers for Android Logic Bombs, the Network and Distributed System Security (NDSS) Symposium 2026, February 24th, 2026, San Diego, CA, USA
- A Novel Fully Sensor-driven Attack Chain, Annual Computer Security Applications Conference (ACSAC) poster session, December 11th, 2025, Honolulu, Hawaii, USA
- Invited talk: A novel fully sensor-driven attack chain, I2S Student Organization (ISO) Meeting, October 23rd, 2025, Nichols Hall, The University of Kansas.
- Agile: An Effective Laser-Based Physical Adversarial Attack against Face Recognition, AI summit *Falling into AI* with Google, October 1st, 2025, Nichols Hall, The University of Kansas.
- The Invisible Polyjuice Potion: An Effective Physical Adversarial Attack against Face Recognition, ACM SIGSAC Conference on Computer and Communications Security (CCS '24), October 17th, 2024, Salt Lake City.
- The Invisible Polyjuice Potion: An Effective Physical Adversarial Attack against Face Recognition, The Central Area Networking and Security Workshop (CANSec) 2024, October 12th, 2024, University of Oklahoma, Norman, OK.
- Stealthily Evading Surveillance Camera Face Recognition, FBI and KU Cybersecurity Conference, April 4th, 2024, KU Memorial Union, the University of Kansas.
- Laser Man against Unauthorized Facial Recognition Systems, The I2S Student Research Symposium (ISRS), March 3rd, 2023, Nichols Hall, The University of Kansas.

Professional Memberships

- ACM SIGSAC membership
- Internet Society membership

Professional Service

Paper Review

- External paper reviewer: USENIX Security '26
- Journal reviewer for Neural Networks
- Journal reviewer for IEEE Transactions on Dependable and Secure Computing (TDSC)
- External paper reviewer: ACM SIGSAC Conference on Computer and Communications Security (CCS 2025)
- External paper reviewer: the 44th IEEE International Conference on Distributed Computing Systems (ICDCS 2024)
- External paper reviewer: Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2023)
- External paper reviewer: the Annual Computer Security Applications Conference (ACSAC 2023)
- External paper reviewer: Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2022)
- External paper reviewer: IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom 2022)

Community Service

- GenCyber Teacher Camp, Student Volunteer and Teaching Assistant, Funded by NSA and NSF, 2023
- GEA Research Symposium, Student Judge, University of Kansas, 2023
- Session Moderator for EAI SecureComm, 2022

References

- **Dr. Fenjun Li** — *EECS, University of Kansas*
  fli@ku.edu — Deane E. Ackers Professor
- **Dr. Bo Luo** — *EECS, University of Kansas*
  bluo@ku.edu — H.J. and Joan O. Wertz Professor
- **Dr. Rongqing Hui** — *EECS, University of Kansas*
  rhui@ku.edu — Professor, Fellow of Optica Society

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